Weak-star and UEB topologies agree on Haar-density probability measures

Candidate full solution for arXiv:2412.00438

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Status. Full solution, likely valid, pending human review.

1 Source and open question

The source is V. Alekseev, H. Ando, F. M. Schneider, and A. Thom, *Amenability and skew-amenability of actions of topological groups*, arXiv:2412.00438v2 (24 October 2025) [1]. After Remark 3.12 on page 9, the authors ask whether, for a locally compact group G with left Haar measure ν, the topology

$$\sigma(C_0(G)^*, C_0(G))$$

on the probability densities

$$\mathcal{P}(G,\nu) = \{\varphi \in L^1(G,\nu) : \varphi \geq 0,\ \|\varphi\|_1 = 1\}$$

agrees with the topology inherited from the UEB topology on $M_u(G_{\mathrm{R}})$, and likewise on $M_u(G_{\mathrm{L}})$. Equivalently, they ask whether the integration maps Ξ_ν^{R} and Ξ_ν^{L} are weak-star-to-UEB continuous.

> We do not know if, for a locally compact group G equipped with a left Haar measure ν, the $\sigma(C_0(G)^*, C_0(G))$-topology on $\mathrm{P}(G,\nu)$ agrees with the UEB topology inherited from $\mathrm{M}_u(G_{\Rsh})$ (or $\mathrm{M}_u(G_{\Lsh})$, resp.). In turn, it is unclear whether $\Xi_\nu^{\Rsh}$ (resp., $\Xi_\nu^{\Lsh}$) is $\sigma(C_0(G)^*, C_0(G))$-to-UEB continuous. However, we will see that for the definition of the AD-amenability of an action, the difference of these topologies is inessential. The following proposition is a corollary to [3, Proposition 2.2] and will be used later in both Theorem 4.5 and Corollary 4.6.

Figure 1: The open question in arXiv:2412.00438v2, page 9.

The source's Appendix Lemma A.2 proves the needed equivalence for maps whose entire family of supports has relatively compact union. The argument below removes that support assumption by deriving a common compact concentration set locally from the weak-star topology itself.

2 Definitions

We identify $\varphi \in \mathcal{P}(G,\nu)$ with the Radon probability measure $\varphi\, d\nu$. A family B of bounded continuous functions on a uniform space is *UEB* if it is uniformly bounded and uniformly equicontinuous. On the dual of the corresponding bounded uniformly continuous functions, the UEB topology is generated by

$$p_B(\lambda) = \sup_{f\in B} |\lambda(f)|$$

as B ranges over UEB families. We apply this definition to the right and left uniformities of G.

We use two standard facts. First,

$$C_0(G) \subseteq \mathrm{RUCB}(G) \cap \mathrm{LUCB}(G). \tag{1}$$

Indeed, compactly supported continuous functions are uniformly continuous for both group uniformities, and $C_0(G)$ is their uniform closure. Second, the restriction of a UEB family to a compact subset $L \subseteq G$ is relatively compact, hence totally bounded, in $C(L)$ by the compact-space Arzela-Ascoli theorem [2].

3 Proof intuition

The apparent mismatch is that weak-star convergence tests one function at a time, whereas a UEB seminorm tests a whole equicontinuous family at once. Probability mass prevents escape at a weak-star limit which is still a probability measure. Near a fixed probability μ, a single compactly supported cutoff forces every nearby probability to put almost all of its mass in one compact set. On that compact set, Arzela-Ascoli reduces a UEB family to finitely many test functions up to a small uniform error. The cutoff turns those finitely many tests into elements of $C_0(G)$, exactly the functions controlled by the weak-star topology. The reverse implication is immediate from (1).

4 The full result

Theorem 1. *Let G be a locally compact Hausdorff group and let ν be a left Haar measure. On $\mathcal{P}(G,\nu)$, the topology $\sigma(C_0(G)^*, C_0(G))$ agrees with the topology pulled back by Ξ^{R}_ν from the UEB topology on $M_u(G_{\mathrm{R}})$. It also agrees with the topology pulled back by Ξ^{L}_ν from the UEB topology on $M_u(G_{\mathrm{L}})$.*

Consequently, both Ξ^{R}_ν and Ξ^{L}_ν are $\sigma(C_0(G)^*, C_0(G))$-to-UEB continuous. Thus the question after Remark 3.12 of arXiv:2412.00438v2 has an affirmative answer, without a second-countability assumption.

Proof. The same argument applies to the right and left uniformities. Fix one of them, write Ξ for the associated integration map, and let B be a UEB family in the associated function algebra. Put

$$M = \sup_{f\in B} \|f\|_\infty < \infty.$$

If $M = 0$, there is nothing to prove. Fix $\mu \in \mathcal{P}(G,\nu)$ and $\varepsilon > 0$. Choose positive numbers η, δ such that

$$3M\eta + 3\delta < \varepsilon. \tag{2}$$

The measure μ is a Radon probability measure. By inner regularity and the locally compact Urysohn lemma, there is a function $\chi \in C_c(G)$ such that

$$0 \le \chi \le 1, \qquad \mu(\chi) > 1 - \eta.$$

Let $L = \operatorname{supp} \chi$, which is compact. The restricted family $B|_L$ is uniformly bounded and equicontinuous. Arzela-Ascoli therefore gives $f_1, \ldots, f_n \in B$ such that for every $f \in B$ some j satisfies

$$\sup_{x \in L} |f(x) - f_j(x)| < \delta. \tag{3}$$

Each χf_j belongs to $C_c(G) \subseteq C_0(G)$.

Consider the following weak-star neighborhood of μ inside $\mathcal{P}(G, \nu)$:

$$V = \{\lambda \in \mathcal{P}(G, \nu) : |\lambda(\chi) - \mu(\chi)| < \eta, \quad |\lambda(\chi f_j) - \mu(\chi f_j)| < \delta \ (1 \le j \le n)\}.$$

For $\lambda \in V$, positivity and normalization give

$$\lambda(1 - \chi) < 2\eta, \qquad \mu(1 - \chi) < \eta.$$

Given $f \in B$, choose f_j as in (3). Splitting with the cutoff and using that χ is supported in L, we obtain

$$\begin{aligned} |\lambda(f) - \mu(f)| &\le |\lambda((1 - \chi)f)| + |\mu((1 - \chi)f)| \\ &\quad + |\lambda(\chi(f - f_j))| + |\mu(\chi(f - f_j))| \\ &\quad + |\lambda(\chi f_j) - \mu(\chi f_j)| \\ &< 3M\eta + 3\delta < \varepsilon. \end{aligned}$$

The bound is uniform over $f \in B$. Hence

$$\sup_{f \in B} |\Xi(\lambda)(f) - \Xi(\mu)(f)| < \varepsilon \qquad (\lambda \in V).$$

Every UEB seminorm is therefore continuous for the weak-star topology on $\mathcal{P}(G, \nu)$. This proves that the weak-star topology is at least as strong as the pulled-back UEB topology.

Conversely, take $h \in C_0(G)$. By (1), the singleton $\{h\}$ is a UEB family for either group uniformity. Moreover,

$$\Xi(\lambda)(h) = \lambda(h) \qquad (\lambda \in \mathcal{P}(G, \nu)).$$

Thus every weak-star coordinate is continuous for the pulled-back UEB topology. The two topologies agree. Applying the argument once to $\mathrm{RUCB}(G)$ and once to $\mathrm{LUCB}(G)$ proves both assertions. $\square$

5 Verification notes

The proof works directly with neighborhoods, so there is no sequentiality or metrizability assumption hidden in the argument. Equivalently, for a net $\mu_i \to \mu$ in the weak-star topology within the probability subspace, the cutoff χ shows eventual uniform tightness of the net; the finite-net estimate then yields convergence in every UEB seminorm.

The normalization to probability measures is essential to the tightness step. In the full positive cone of $C_0(G)^*$, vague convergence can lose mass at infinity (for example, escaping point masses), so the corresponding statement must not be asserted there. No such escape is possible when both the approximants and their weak-star limit lie in $\mathcal{P}(G, \nu)$.

The proof in fact applies to all Radon probability measures on G, not only those absolutely continuous with respect to Haar measure. The narrower formulation above exactly answers the source question.

6 Novelty check and limitations

On 9 August 2026, the run's registry, solution, attempt, and proof-gap indexes were searched for arXiv:2412.00438, the paper title, and the core phrase "weak-star UEB probability topology"; no duplicate was found. A bounded external search used the exact quoted sentence, the displayed $\sigma(C_0(G)^*, C_0(G))$ notation, and close variants involving weak convergence and UEB topology. It found the source paper and general UEB background, but no later paper explicitly resolving this question. The current arXiv v2, dated 24 October 2025, still contains the question on page 9.

This packet answers only that topology/continuity question. It does not claim to settle the paper's separate questions about non-second-countable amenable actions, well-placed subgroups, or extensions of exact groups.

7 Human-review recommendation

Recommended for expert verification and promotion. The reviewer should check: (i) the standard inclusion (1) under the source's left/right uniformity conventions; (ii) the compact Hausdorff form of Arzela-Ascoli used to obtain the finite net; and (iii) the three error terms in the cutoff estimate. No computational or unproved dependency remains.

References

[1] V. Alekseev, H. Ando, F. M. Schneider, and A. Thom, *Amenability and skew-amenability of actions of topological groups*, arXiv:2412.00438v2, 2025.

[2] J. L. Kelley, *General Topology*, Graduate Texts in Mathematics 27, Springer, 1975 reprint; compact-space Arzela-Ascoli theorem.